

The first step is to factor $Q(s)$ into first-order and quadratic factors with real coefficients:

$$F(s) = \frac{P(s)}{Q(s)} = \frac{P(s)}{(s - s_1)(s - s_2) \cdots (s - s_k) \cdots (s - s_n)} \quad (4.33)$$

The values $s_1, s_2, \dots, s_n$ in the finite plane that make the denominator equal to zero are called the *zeros* of the denominator. These values of s , which may be either real or complex, also make $|F(s)|$ infinite, and so they are called *poles* of $F(s)$. Therefore, the values $s_1, s_2, \dots, s_n$ are referred to as zeros of the denominator or poles of the complete function in the finite plane, i.e., there are n poles of $F(s)$. Methods of factoring polynomials exist in the literature. Digital-computer programs are available to perform this operation (see [Appendix C](#)) [4].

The transform $F(s)$ can be expressed as a series of fractions. If the poles are simple (nonrepeated), the number of fractions is equal to n , the number of poles of $F(s)$. In such case the function $F(s)$ can be expressed as

$$F(s) = \frac{P(s)}{Q(s)} = \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} + \cdots + \frac{A_k}{s - s_k} + \cdots + \frac{A_n}{s - s_n} \quad (4.34)$$

The procedure is to evaluate the constants $A_1, A_2, \dots, A_n$ corresponding to the poles $s_1, s_2, \dots, s_n$. The coefficients $A_1, A_2, \dots$ are termed the *residues*[†] of $F(s)$ at the corresponding poles. Cases of repeated factors and complex factors are treated separately. Several ways of evaluating the constants are shown in the following section.

4.8 HEAVISIDE PARTIAL-FRACTION EXPANSION THEOREMS

The technique of partial-fraction expansion is set up to take care of all cases systematically. There are four classes of problems, depending on the denominator $Q(s)$. Each of these cases is illustrated separately.

- Case 1* $F(s)$ has first-order real poles.
- Case 2* $F(s)$ has repeated first-order real poles.
- Case 3* $F(s)$ has a pair of complex-conjugate poles (a quadratic factor in the denominator).
- Case 4* $F(s)$ has repeated pairs of complex-conjugate poles (a repeated quadratic factor in the denominator).

[†]More generally the residue is the coefficient of the $(s - s_i)^{-1}$ term in the Laurent expansion of $F(s)$ about $s = s_i$.

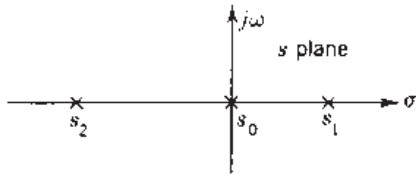


FIGURE 4.1 Location of real poles in the s plane.

Case 1: First-Order Real Poles

The locations of three real poles of $F(s)$ in the s plane are shown in Fig. 4.1. The poles may be positive, zero, or negative, and they lie on the real axis in the s plane. In this example, s_1 is positive, s_0 is zero, and s_2 is negative. For the poles shown in Fig. 4.1 the transform $F(s)$ and its partial fractions are

$$F(s) = \frac{P(s)}{Q(s)} = \frac{P(s)}{s(s - s_1)(s - s_2)} = \frac{A_0}{s} + \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} \quad (4.35)$$

There are as many fractions as there are factors in the denominator of $F(s)$. Since $s_0 = 0$, the factor $s - s_0$ is written simply as s . The inverse transform of $F(s)$ is

$$f(t) = A_0 + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (4.36)$$

The pole s_1 is positive; therefore, the term $A_1 e^{s_1 t}$ is an increasing exponential and the system is unstable. The pole s_2 is negative, and the term $A_2 e^{s_2 t}$ is a decaying exponential with a final value of zero. Therefore, for a system to be stable, all real poles that contribute to the complementary solution must be in the left half of the s plane.

To evaluate a typical coefficient A_k , multiply both sides of Eq. (4.34) by the factor $s - s_k$. The result is

$$\begin{aligned} (s - s_k)F(s) &= (s - s_k) \frac{P(s)}{Q(s)} \\ &= A_1 \frac{s - s_k}{s - s_1} + A_2 \frac{s - s_k}{s - s_2} + \cdots + A_k + \cdots + A_n \frac{s - s_k}{s - s_n} \end{aligned} \quad (4.37)$$

The multiplying factor $s - s_k$ on the left side of the equation and the same factor of $Q(s)$ should be canceled. By letting $s = s_k$, each term on the right side of the equation is zero except A_k . Thus, a general rule for evaluating the constants for single-order real poles is

$$A_k = \left[(s - s_k) \frac{P(s)}{Q(s)} \right]_{s=s_k} = \left[\frac{P(s)}{Q'(s)} \right]_{s=s_k} \quad (4.38)$$

where $Q'(s_k) = [dQ(s)/ds]_{s=s_k} = [Q(s)/(s-s_k)]_{s=s_k}$. The coefficients A_k are the *residues* of $F(s)$ at the corresponding poles. For the case of

$$F(s) = \frac{s+2}{s(s+1)(s+3)} = \frac{A_0}{s} + \frac{A_1}{s+1} + \frac{A_2}{s+3}$$

the constants are

$$A_0 = [sF(s)]_{s=0} = \left[\frac{s+2}{(s+1)(s+3)} \right]_{s=0} = \frac{2}{3}$$

$$A_1 = [(s+1)F(s)]_{s=-1} = \left[\frac{s+2}{s(s+3)} \right]_{s=-1} = -\frac{1}{2}$$

$$A_2 = [(s+3)F(s)]_{s=-3} = \left[\frac{s+2}{s(s+1)} \right]_{s=-3} = -\frac{1}{6}$$

The solution as a function of time is

$$f(t) = \frac{2}{3} - \frac{1}{2}e^{-t} - \frac{1}{6}e^{-3t} \quad (4.39)$$

Case 2: Multiple-Order Real Poles

The position of real poles of $F(s)$, some of which are repeated, is shown in Fig. 4.2. The notation $]_r$ indicates a pole of order r . All real poles lie on the real axis of the s plane. For the poles shown in Fig. 4.2 the transform $F(s)$ and its partial fractions are

$$\begin{aligned} F(s) &= \frac{P(s)}{Q(s)} = \frac{P(s)}{(s-s_1)^3(s-s_2)} \\ &= \frac{A_{13}}{(s-s_1)^3} + \frac{A_{12}}{(s-s_1)^2} + \frac{A_{11}}{s-s_1} + \frac{A_2}{s-s_2} \end{aligned} \quad (4.40)$$

The order of $Q(s)$ in this case is 4, and there are four fractions. Note that the multiple pole s_1 , which is of order 3, has resulted in three fractions on the right

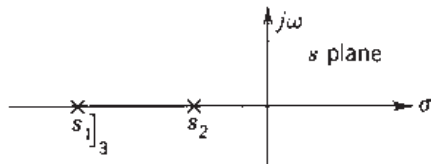


FIGURE 4.2 Location of real poles in the s plane.

side of Eq. (4.40). To designate the constants in the partial fractions, a single subscript is used for a first-order pole. For multiple-order poles a double-subscript notation is used. The first subscript designates the pole, and the second subscript designates the order of the pole in the partial fraction. The constants associated with first-order denominators in the partial-fraction expansion are called residues: therefore, only the constants A_{11} and A_2 are residues of Eq. (4.40).

The inverse transform of $F(s)$ (see transform pairs 7 and 8 in [Appendix A](#)) is

$$f(t) = A_{13} \frac{t^2}{2} e^{s_1 t} + A_{12} t e^{s_1 t} + A_{11} e^{s_1 t} + A_2 e^{s_2 t} \quad (4.41)$$

For the general transform with repeated real roots,

$$\begin{aligned} F(s) &= \frac{P(s)}{Q(s)} = \frac{P(s)}{(s - s_q)^r (s - s_1) \cdots} \\ &= \frac{A_{qr}}{(s - s_q)^r} + \frac{A_{q(r-1)}}{(s - s_q)^{r-1}} + \cdots + \frac{A_{q(r-k)}}{(s - s_q)^{r-k}} + \cdots \\ &\quad + \frac{A_{q1}}{s - s_q} + \frac{A_1}{s - s_1} + \cdots \end{aligned} \quad (4.42)$$

The constant A_{qr} can be evaluated by simply multiplying both sides of Eq. (4.42) by $(s - s_q)^r$, giving

$$\begin{aligned} (s - s_q)^r F(s) &= \frac{(s - s_q)^r P(s)}{Q(s)} = \frac{P(s)}{(s - s_1) \cdots} \\ &= A_{qr} + A_{q(r-1)}(s - s_q) + \cdots \\ &\quad + A_{q1}(s - s_q)^{r-1} + A_1 \frac{(s - s_q)^r}{s - s_1} + \cdots \end{aligned} \quad (4.43)$$

Note that the factor $(s - s_q)^r$ must be divided out of the left side of the equation.

For $s = s_q$, all terms on the right side of the equation are zero except A_{qr} ; therefore,

$$A_{qr} = \left[(s - s_q)^r \frac{P(s)}{Q(s)} \right]_{s=s_q} \quad (4.44)$$

Evaluation of $A_{q(r-1)}$ cannot be performed in a similar manner. Multiplying both sides of Eq. (4.42) by $(s - s_q)^{r-1}$ and letting $s = s_q$ would result in both sides being infinite, which leaves $A_{q(r-1)}$ indeterminate. If the term A_{qr} were eliminated from Eq. (4.43), $A_{q(r-1)}$ could be evaluated. This can be done by differentiating Eq. (4.43) with respect to s :

$$\frac{d}{ds} \left[(s - s_q)^r \frac{P(s)}{Q(s)} \right] = A_{q(r-1)} + 2A_{q(r-2)}(s - s_q) + \cdots \quad (4.45)$$

Letting $s = s_q$ gives

$$A_{q(r-1)} = \left\{ \frac{d}{ds} \left[(s - s_q)^r \frac{P(s)}{Q(s)} \right] \right\}_{s=s_q} \quad (4.46)$$

Repeating the differentiation gives the coefficient $A_{q(r-2)}$ as

$$A_{q(r-2)} = \left\{ \frac{1}{2} \frac{d^2}{ds^2} \left[(s - s_q)^r \frac{P(s)}{Q(s)} \right] \right\}_{s=s_q} \quad (4.47)$$

This process can be repeated until each constant is determined. A general formula for finding these coefficients associated with the repeated real pole of order r is

$$A_{q(r-k)} = \left\{ \frac{1}{k!} \frac{d^k}{ds^k} \left[(s - s_q)^r \frac{P(s)}{Q(s)} \right] \right\}_{s=s_q} \quad (4.48)$$

For the case of

$$F(s) = \frac{1}{(s+2)^3(s+3)} = \frac{A_{13}}{(s+2)^3} + \frac{A_{12}}{(s+2)^2} + \frac{A_{11}}{s+2} + \frac{A_2}{s+3} \quad (4.49)$$

the constants are

$$\begin{aligned} A_{13} &= [(s+2)^3 F(s)]_{s=-2} = 1 & A_{12} &= \left\{ \frac{d}{ds} [(s+2)^3 F(s)] \right\}_{s=-2} = -1 \\ A_{11} &= \left\{ \frac{d^2}{ds^2} [(s+2)^3 F(s)] \right\}_{s=-2} = 1 & A_2 &= [(s+3)F(s)]_{s=-3} = -1 \end{aligned}$$

and the solution as a function of time is

$$f(t) = \frac{t^2}{2} e^{-2t} - t e^{-2t} + e^{-2t} - e^{-3t} \quad (4.50)$$

A recursive algorithm for computing the partial fraction expansion of rational functions having any number of multiple poles is readily programmed on a digital computer and is given in Ref. 5.

Case 3: Complex-Conjugate Poles

The position of complex poles of $F(s)$ in the s plane is shown in Fig. 4.3. Complex poles always are present in complex-conjugate pairs; their real part may be either positive or negative. For the poles shown in Fig. 4.3 the

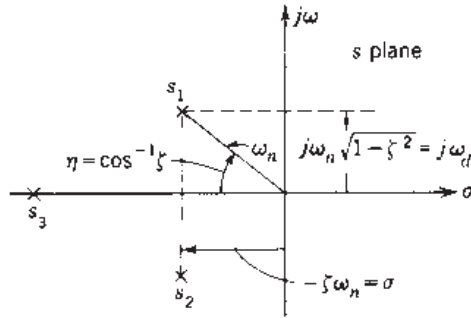


FIGURE 4.3 Location of complex-conjugate poles in the s plane.

transform $F(s)$ and its partial fractions are

$$\begin{aligned}
 F(s) &= \frac{P(s)}{Q(s)} = \frac{P(s)}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(s - s_3)} \\
 &= \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} + \frac{A_3}{s - s_3} \\
 &= \frac{A_1}{s + \zeta\omega_n - j\omega_n\sqrt{1 - \zeta^2}} + \frac{A_2}{s + \zeta\omega_n + j\omega_n\sqrt{1 - \zeta^2}} + \frac{A_3}{s - s_3} \quad (4.51)
 \end{aligned}$$

The inverse transform of $F(s)$ is

$$\begin{aligned}
 f(t) &= A_1 \exp \left[\left(-\zeta\omega_n + j\omega_n\sqrt{1 - \zeta^2} \right) t \right] \\
 &\quad + A_2 \exp \left[\left(-\zeta\omega_n - j\omega_n\sqrt{1 - \zeta^2} \right) t \right] + A_3 e^{s_3 t} \quad (4.52)
 \end{aligned}$$

Since the poles s_1 and s_2 are complex conjugates, and since $f(t)$ is a real quantity, the coefficients A_1 and A_2 must also be complex conjugates. Equation (4.52) can be written with the first two terms combined to a more useful damped sinusoidal form:

$$\begin{aligned}
 f(t) &= 2|A_1|e^{-\zeta\omega_n t} \sin \left(\omega_n\sqrt{1 - \zeta^2}t + \phi \right) + A_3 e^{s_3 t} \\
 &= 2|A_1|e^{\sigma t} \sin (\omega_d t + \phi) + A_3 e^{s_3 t} \quad (4.53)
 \end{aligned}$$

where $\phi = \text{angle of } A_1 + 90^\circ$, $\omega_d = \omega_n\sqrt{1 - \zeta^2}$, and $\sigma = -\zeta\omega_n$.

The values of A_1 and A_3 , as found in the manner shown previously, are

$$A_1 = [(s - s_1)F(s)]_{s=s_1} \quad \text{and} \quad A_3 = [(s - s_3)F(s)]_{s=s_3}$$

Since s_1 is complex, the constant A_1 is also complex. *Remember that A_1 is associated with the complex pole with the positive imaginary part.*

In Fig. 4.3 the complex poles have a negative real part, $\sigma = -\zeta\omega_n$, where the damping ratio ζ is positive. For this case the corresponding transient response is known as a damped sinusoid and is shown in Fig. 3.2. Its

final value is zero. The angle η shown in Fig. 4.3 is measured from the negative real axis and is related to the damping ratio by

$$\cos \eta = \zeta$$

If the complex pole has a positive real part, the time response increases exponentially with time and the system is unstable. If the complex roots are in the right half of the s plane, the damping ratio ζ is negative. The angle η for this case is measured from the positive real axis and is given by

$$\cos \eta = |\zeta|$$

For the case of

$$F(s) = \frac{10}{(s^2 + 6s + 25)(s + 2)} = \frac{A_1}{s + 3 - j4} + \frac{A_2}{s + 3 + j4} + \frac{A_3}{s + 2} \quad (4.54)$$

the constants are

$$\begin{aligned} A_1 &= \left[(s + 3 - j4) \frac{10}{(s^2 + 6s + 25)(s + 2)} \right]_{s=-3+j4} \\ &= \left[\frac{10}{(s + 3 + j4)(s + 2)} \right]_{s=-3+j4} = 0.303 \angle -194^\circ. \\ A_3 &= \left[(s + 2) \frac{10}{(s^2 + 6s + 25)(s + 2)} \right]_{s=-2} = \left(\frac{10}{s^2 + 6s + 25} \right)_{s=-2} = 0.59 \end{aligned}$$

Using Eq. (4.53), the solution is

$$f(t) = 0.606e^{-3t} \sin(4t - 104^\circ) + 0.59e^{-2t} \quad (4.55)$$

The function $F(s)$ of Eq. (4.54) appears in Appendix A as transform pair 29. With the notation of Appendix A, the phase angle in the damped sinusoidal term is

$$\phi = \tan^{-1} \frac{b}{c-a} = \tan^{-1} \frac{4}{2-3} = \tan^{-1} \frac{4}{-1} = 104^\circ \quad (4.56)$$

It is important to note that $\tan^{-1}[4/(-1)] \neq \tan^{-1}(-4/1)$. To get the correct value for the angle ϕ , it is useful to draw a sketch in order to avoid ambiguity and ensure that ϕ is evaluated correctly.

IMAGINARY POLES. The position of imaginary poles of $F(s)$ in the s plane is shown in Fig. 4.4. As the real part of the poles is zero, the poles lie on the imaginary axis. This situation is a special case of complex poles, i.e., the damping ratio $\zeta = 0$. For the poles shown in Fig. 4.4 the transform $F(s)$ and its partial fractions are

$$F(s) = \frac{P(s)}{Q(s)} = \frac{P(s)}{(s^2 + \omega_n^2)(s - s_3)} = \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} + \frac{A_3}{s - s_3} \quad (4.57)$$

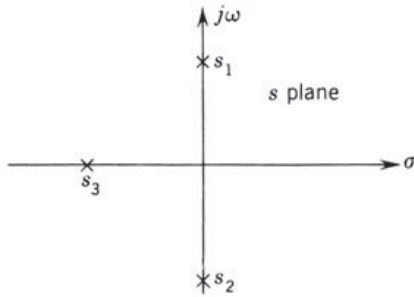


FIGURE 4.4 Poles of $F(s)$ containing imaginary conjugate poles in the s plane.

The quadratic can be factored in terms of the poles s_1 and s_2 ; thus,

$$s^2 + \omega_n^2 = (s - j\omega_n)(s + j\omega_n) = (s - s_1)(s - s_2)$$

The inverse transform of Eq. (4.57) is

$$f(t) = A_1 e^{j\omega_n t} + A_2 e^{-j\omega_n t} + A_3 e^{s_3 t} \quad (4.58)$$

As $f(t)$ is a real quantity, the coefficients A_1 and A_2 are complex conjugates. The terms in Eq. (4.58) can be combined into the more useful form

$$f(t) = 2|A_1| \sin(\omega_n t + \phi) + A_3 e^{s_3 t} \quad (4.59)$$

Note that since there is no damping term multiplying the sinusoid, this term represents a steady-state value. The angle ϕ is given by

$$\phi = \text{angle of } A_1 + 90^\circ$$

The values of A_1 and A_3 , found in the conventional manner, are

$$A_1 = [(s - s_1)F(s)]_{s=s_1} \quad A_3 = [(s - s_3)F(s)]_{s=s_3}$$

For the case where

$$F(s) = \frac{100}{(s^2 + 25)(s + 2)} = \frac{A_1}{s - j5} + \frac{A_2}{s + j5} + \frac{A_3}{s + 2} \quad (4.60)$$

the values of the coefficients are

$$A_1 = [(s - j5)F(s)]_{s=j5} = \left[\frac{100}{(s + j5)(s + 2)} \right]_{s=j5} = 1.86 \angle -158.2^\circ$$

$$A_3 = [(s + 2)F(s)]_{s=-2} = \left(\frac{100}{s^2 + 25} \right)_{s=-2} = 3.45$$

The solution is

$$f(t) = 3.72 \sin(5t - 68.2^\circ) + 3.45 e^{-2t} \quad (4.61)$$

Case 4: Multiple-Order Complex Poles

Multiple-order complex-conjugate poles are rare. They can be treated in much the same fashion as repeated real poles.

4.9 MATLAB PARTIAL-FRACTION EXAMPLE

The MATLAB *RESIDUE* command is used to obtain the partial-fraction expansion of Eq. (4.54).

MATLAB Partial-Fraction m-file

```
%Set up transfer function
%
%   Define numerator constant of F(s)
k=10;
%   Define poles of F(s)
p(1)=-3+j*4; p(2)=-3-j*4; p(3)=-2;
%Define numerator and denominator
num=k;
den=poly(p)
den=
1     8     37     50
%
%   Use the RESIDUE command to call for residues:
[r1, p1, k1=residue (num, den);
%   List the residues (which are complex)
r1
r1=
-0.2941+0.0735i
-0.2941-0.0735i
0.5882
%   List the poles to check data entry
p1
p1=
-3.0000+4.0000i
-3.0000-4.0000i
-2.0000
%   List the angle of the residues in degrees,
angle (r1)*180/pi
ans=
165.9638
-165.9638
0
%   List the magnitudes of the residues
abs (r1)
```

```

ans =
0.3032
0.3032
0.5882
%
% To show numerical stability, form the transfer function from
% the residues (CADAC) (accuracy check)
% CADAC (Accuracy check)
[num1, den1]=residue (r1, p1, k1);
printsys (num1, den1)
num/den=

1.1102e - 016s^2 + 4.4409e - 016s + 10
-----
s^3 + 8s^2 + 37s + 50

%
% Note the false roots in the numerator
% Form the Poles
roots (den1)
ans =
-3.0000 + 4.0000i
-3.0000 - 4.0000i
-2.0000
%
% Plot a three second response to an impulse
% Define a time vector [initial value: incremental value:
% final value]
time = [0: .02: 3];
% Use impulse response command
impulse (num, den, time)
% Print the plot to compare it with the plot in Fig. 4.5
print -dtiff Sec.4.9. tif

```

The PARTFRAC Command

The m-file name *partfrac.m* calculates time responses and is included on the accompanying CDROM.

```
Partfrac (num, den)
```

MATLAB produces the following output, which corresponds to Eq. (4.55).

```
The partial fraction expansion of
Num/den=
```

$$\frac{10}{s^3 + 8s^2 + 37s + 50}$$

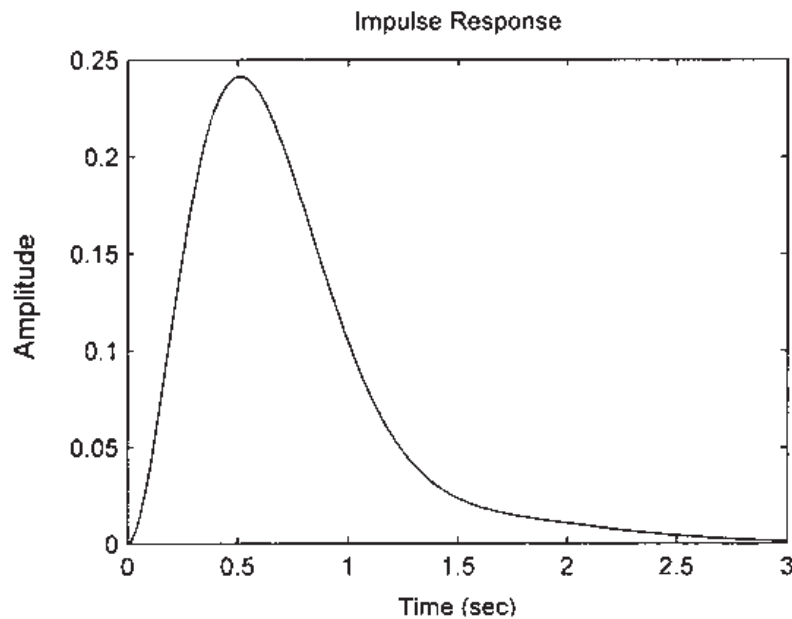


FIGURE 4.5 Response of Eq. (4.55) from MATLAB.

Representing the time response for a unit impulse input is
 $y(t) =$

$$0.60634 \exp(-3t) \cos(4t + 165.9638\text{deg}) + 0.58824 \exp(-2t)$$

4.10 PARTIAL-FRACTION SHORTCUTS

Some shortcuts can be used to simplify the partial-fraction expansion procedures. They are most useful for transform functions that have multiple poles or complex poles. With multiple poles, the evaluation of the constants by the process of repeated differentiation given by Eq. (4.48) can be very tedious, particularly when a factor containing s appears in the numerator of $F(s)$. The modified procedure is to evaluate only those coefficients for real poles that can be obtained readily without differentiation. The corresponding partial functions are then subtracted from the original function. The resultant function is easier to work with to get the additional partial fractions than the original function $F(s)$.

Example 1. For the function $F(s)$ of Eq. (4.49), which contains a pole $s_1 = -2$ of multiplicity 3, the coefficient $A_{13} = 1$ is readily found. Subtracting the associated fraction from $F(s)$ and putting the result over a common

denominator yields.

$$F_x(s) = \frac{1}{(s+2)^3(s+3)} - \frac{1}{(s+2)^3} = \frac{-1}{(s+2)^2(s+3)}$$

Note that the remainder $F_x(s)$ contains the pole $s_1 = -2$ with a multiplicity of only 2. Thus the coefficient A_{12} is now easily obtained from $F_x(s)$ without differentiation. The procedure is repeated to obtain all the coefficients associated with the multiple pole.

Example 2. Consider the function

$$F(s) = \frac{20}{(s^2 + 6s + 25)(s+1)} = \frac{As + B}{s^2 + 6s + 25} + \frac{C}{s+1} \quad (4.62)$$

which has one real pole and a pair of complex poles. Instead of partial fractions for each pole, leave the quadratic factor in one fraction. Note that the numerator of the fraction containing the quadratic is a polynomial in s and is one degree lower than the denominator. Residue C is $C = [(s+1)F(s)]_{s=-1} = 1$. Subtracting this fraction from $F(s)$ gives

$$\frac{20}{(s^2 + 6s + 25)(s+1)} - \frac{1}{s+1} = \frac{-s^2 - 6s - 5}{(s^2 + 6s + 25)(s+1)}$$

Since the pole at $s = -1$ has been removed, the numerator of the remainder must be divisible by $s+1$. The simplified function must be equal to the remaining partial fraction; i.e.,

$$-\frac{s+5}{s^2 + 6s + 25} = \frac{As + B}{s^2 + 6s + 25} = \frac{-(s+5)}{(s+3)^2 + 4^2}$$

The inverse transform for this fraction can be obtained directly from [Appendix A](#), pair 26. The complete inverse transform of $F(s)$ is

$$f(t) = e^{-t} - 1.12e^{-3t} \sin(4t + 63.4^\circ) \quad (4.63)$$

Leaving the complete quadratic in one fraction has resulted in real constants for the numerator. The chances for error are probably less than evaluating the residues, which are complex numbers, at the complex poles.

A rule presented by Hazony and Riley [6] is very useful for evaluating the coefficients of partial-fraction expansions. For a normalized[‡] ratio of polynomials the rule is expressed as follows:

1. If the denominator is 1 degree higher than the numerator, the sum of the residues is 1.

[‡]A normalized polynomial is a monic polynomial, that is, the coefficient of the term containing the highest degree is equal to unity.

2. If the denominator is 2 or more degrees higher than the numerator, the sum of the residues is 0.

Equation (4.32) with a_w factored from the numerator is a normalized ratio of polynomials. These rules are applied to the ratio $F(s)/a_w$. It should be noted that *residue* refers only to the coefficients of terms in a partial-fraction expansion with first-degree denominators. Coefficients of terms with higher-degree denominators are referred to only as coefficients. These rules can be used to simplify the work involved in evaluating the coefficients of partial-fraction expansions, particularly when the original function has a multiple-order pole. For example, only A_{11} and A_2 in Eq. (4.49) are residues, and therefore $A_{11} + A_2 = 0$. Since $A_2 = -1$, the value of $A_{11} = 1$ is obtained directly.

Digital-computer programs (see [Appendixes C and D](#)) are readily available for evaluating the partial-fraction coefficients and obtaining a tabulation and plot of $f(t)$ [4].

4.11 GRAPHICAL INTERPRETATION OF PARTIAL-FRACTION COEFFICIENTS [7]

The preceding sections describe the analytical evaluation of the partial-function coefficients. These constants are directly related to the pole-zero pattern of the function $F(s)$ and can be determined graphically, whether the poles and zeros are real or in complex-conjugate pairs. As long as $P(s)$ and $Q(s)$ are in factored form, the coefficients can be determined graphically by inspection. Rewriting Eq. (4.32) with the numerator and denominator in factored form and with $a_w = K$ gives

$$\begin{aligned} F(s) &= \frac{P(s)}{Q(s)} = \frac{K(s - z_1)(s - z_2) \cdots (s - z_m) \cdots (s - z_w)}{(s - p_1)(s - p_2) \cdots (s - p_k) \cdots (s - p_n)} \\ &= K \frac{\prod_{m=1}^w (s - z_m)}{\prod_{k=1}^n (s - p_k)} \end{aligned} \quad (4.64)$$

The zeros of this function, $s = z_m$, are those values of s for which the function is zero; that is, $F(z_m) = 0$. Zeros are indicated by a small circle on the s plane. The poles of this function, $s = p_k$, are those values of s for which the function is infinite; that is, $|F(p_k)| = \infty$. Poles are indicated by a small $\times$ on the s plane. Poles are also known as singularities[§] of the function. When $F(s)$ has only simple poles (first-order poles), it can be expanded into partial fractions of

[§]A singularity of a function $F(s)$ is a point where $F(s)$ does not have a derivative.

the form

$$F(s) = \frac{A_1}{s - p_1} + \frac{A_2}{s - p_2} + \cdots + \frac{A_k}{s - p_k} + \cdots + \frac{A_n}{s - p_n} \quad (4.65)$$

The coefficients are obtained from Eq. (4.38) and are given by

$$A_k = [(s - p_k)F(s)]_{s=p_k} \quad (4.66)$$

The first coefficient is

$$A_1 = \frac{K(p_1 - z_1)(p_1 - z_2) \cdots (p_1 - z_w)}{(p_1 - p_2)(p_1 - p_3) \cdots (p_1 - p_n)} \quad (4.67)$$

Figure 4.6 shows the poles and zeros of a function $F(s)$. The quantity $s = p_1$ is drawn as an arrow from the origin of the s plane to the pole p_1 . This arrow is called a *directed line segment*. It has a magnitude equal to $|p_1|$ and an angle of 180° . Similarly, the directed line segment $s = z_1$ is drawn as an arrow from the origin to the zero z_1 and has a corresponding magnitude and angle. A directed line segment is also drawn from the zero z_1 to the pole p_1 . By the rules of vector addition, this directed line segment is equal to $p_1 - z_1$; in polar form it has a magnitude equal to its length and an angle ψ , as shown. By referring to Eq. (4.67), it is seen that $p_1 - z_1$ appears in the numerator of A_1 . Whereas this quantity can be evaluated analytically, it can also be measured from the pole-zero diagram. In a similar fashion, each factor in Eq. (4.67) can be obtained graphically; the directed line segments are drawn *from* the zeros and each of the other poles *to* the pole p_1 . The angle from a zero is indicated by the symbol ψ , and the angle from a pole is indicated by the symbol θ .

The general rule for evaluating the coefficients in the partial-fraction expansion is quite simple. The value of A_k is the product of K and the directed distances from each zero to the pole p_k divided by the product of the directed distances from each of the other poles to the pole p_k . Each of these

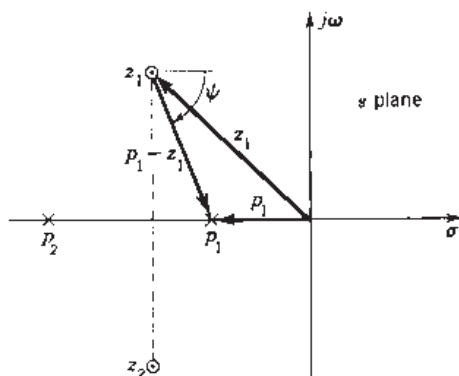


FIGURE 4.6 Directed line segments drawn on a pole-zero diagram of a function $F(s)$.

directed distances is characterized by a magnitude and an angle. This statement can be written in equation form as

$$A_k = K \frac{\text{product of directed distances from each zero to pole } p_k}{\text{product of directed distances from all other poles to pole } p_k}$$

$$K = \frac{\prod_{m=1}^w |p_k - z_m| \angle \sum_{m=1}^w \psi(p_k - z_m)}{\prod_{c=1, c \neq k}^n |p_k - p_c| \angle \sum_{c=1}^n \theta(p_k - p_c)} \quad (4.68)$$

Equations (4.67) and (4.68) readily reveals that as the pole p_k comes closer to the zero z_m , the corresponding coefficient A_k becomes smaller (approaches a zero value). This characteristic is very important in analyzing system response characteristics. For real poles the values of A_k must be real but can be either positive or negative. A_k is positive if the total number of real poles and real zeros to the right of p_k is even, and it is negative if the total number of real poles and real zeros to the right of p_k is odd. For complex poles the values of A_k are complex. If $F(s)$ has no finite zeros, the numerator of Eq. (4.68) is equal to K . A repeated factor of the form $(s + a)^r$ in either the numerator or the denominator is included r times in Eq. (4.68), where the values of $|s + a|$ and $\angle s + a$ are obtained graphically. The application of the graphical technique is illustrated by the following example:

Example. A function $F(s)$ with complex poles (step function response):

$$F(s) = \frac{K(s + \alpha)}{s[(s + a)^2 + b^2]} = \frac{K(s + \alpha)}{s(s + a - jb)(s + a + jb)}$$

$$= \frac{K(s - z_1)}{s(s - p_1)(s - p_2)} = \frac{A_0}{s} + \frac{A_1}{s + a - jb} + \frac{A_2}{s + a + jb} \quad (4.69)$$

The coefficient A_0 , for the pole $p_0 = 0$, is obtained by use of the directed line segments shown in Fig. 4.7a:

$$A_0 = \frac{K(\alpha)}{(a - jb)(a + jb)} = \frac{K\alpha}{\sqrt{a^2 + b^2} e^{j\theta_1} \sqrt{a^2 + b^2} e^{j\theta_2}} = \frac{K\alpha}{a^2 + b^2} \quad (4.70)$$

Note that the angles θ_1 and θ_2 are equal in magnitude but opposite in sign. The coefficient for a real pole is therefore always a real number.

The coefficient A_1 , for the pole $p_1 = -a + jb$, is obtained by use of the directed line segments shown in Fig. 4.7b:

$$A_1 = \frac{K[(\alpha - a) + jb]}{(-a + jb)(j2b)} = \frac{K\sqrt{(\alpha - a)^2 + b^2} e^{j\psi}}{\sqrt{a^2 + b^2} e^{j\theta} 2be^{j\pi/2}}$$

$$= \frac{K}{2b} \sqrt{\frac{(\alpha - a)^2 + b^2}{a^2 + b^2}} e^{j(\psi - \theta - \pi/2)} \quad (4.71)$$

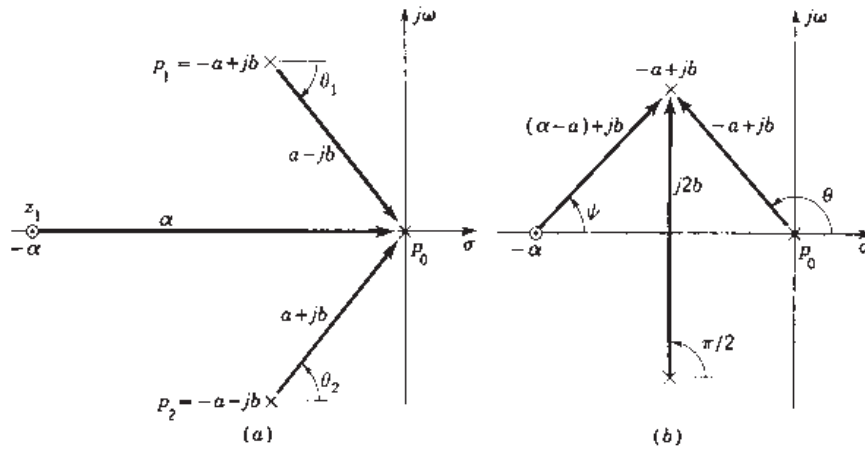


FIGURE 4.7 Directed line segments for evaluating (a) A_0 of Eq. (4.70) and (b) A_1 of Eq. (4.71).

where

$$\psi = \tan^{-1} \frac{b}{\alpha - a} \quad \text{and} \quad \theta = \tan^{-1} \frac{b}{-a}$$

Since the constants A_1 and A_2 are complex conjugates, it is not necessary to evaluate A_2 . It should be noted that zeros in $F(s)$ affect both the magnitude and the angle of the coefficients in the time function.

The response as a function of time is therefore

$$f(t) = A_0 + 2|A_1|e^{-at} \sin(bt + \angle A_1 + 90^\circ)$$

$$= \frac{K\alpha}{a^2 + b^2} + \frac{K}{b} \sqrt{\frac{(\alpha - a)^2 + b^2}{a^2 + b^2}} e^{-at} \sin(bt + \phi) \quad (4.72)$$

where

$$\phi = \psi - \theta = \tan^{-1} \frac{b}{\alpha - a} - \tan^{-1} \frac{b}{-a} = \text{angle of } A_1 + 90^\circ \quad (4.73)$$

This agrees with transform pair 28 in [Appendix A](#). A_0 and A_1 can be evaluated by measuring the directed line segments directly on the pole-zero diagram. It is interesting to note that, for a given ζ , as the magnitude $\omega_n = \sqrt{a^2 + b^2}$ becomes larger, the coefficients A_0 and A_1 approach zero as revealed by an analysis of Eqs. (4.70) and (4.71), respectively. (The limiting value is $\lim_{\omega_n \rightarrow \infty} [A_1] = K/2b$.) This is an important characteristic when it is desired that $f(t)$ be very small. The application of this characteristic